FR3 RViz, Gazebo & Gripper

Practical Guide for Fake-Hardware Visualization, MoveIt Launch Structure, Gazebo Controllers and Gripper Testing

This guide combines the supplied RViz, gripper, and launch-file notes into one cleaned-up English reference.

Workspace	~/franka_ros2_ws
Main MoveIt package	franka_fr3_moveit_config
Gazebo bringup package	franka_gazebo_bringup
Robot	Franka Research 3 (FR3)
Main visualization	RViz / MoveIt
Simulation controllers	Joint position, joint velocity, joint impedance

1. RViz-Only Simulation with Fake Hardware

1.1 Build and launch

The first workflow explicitly rebuilds the FR3 MoveIt configuration package before launching:

```
cd ~/franka_ros2_ws
colcon build --packages-select franka_fr3_moveit_config
source ~/franka_ros2_ws/install/setup.bash

QT_QPA_PLATFORM=xcb ros2 launch franka_fr3_moveit_config moveit.launch.py \
  robot_ip:=dont-care \
  use_fake_hardware:=true
```

If the package has already been built and the workspace is already sourced, the shorter launch command is:

```
ros2 launch franka_fr3_moveit_config moveit.launch.py \
  robot_ip:=dont-care \
  use_fake_hardware:=true
```

> NOTE: The two commands launch the same fake-hardware setup. The first form additionally forces the Qt platform to xcb, which can help when RViz has display-related issues.

1.2 RViz RobotModel display settings

To make the robot model visible in RViz, add a RobotModel display and configure:

- Description Source: Topic
- Description Topic: /robot_description
- Durability Policy: Transient Local

2. Changes Made to moveit.launch.py

The file modified in the notes is:

```
~/franka_ros2_ws/src/franka_fr3_moveit_config/launch/moveit.launch.py
```

The following changes were made so that the setup would work correctly:

- Removed kinematics_yaml from the RViz node configuration.
- Changed the gripper include from Include to IncludeLaunchDescription.
- Added the required launch imports, including LaunchDescription and IncludeLaunchDescription.
- Changed controller startup order so that joint_state_broadcaster starts before fr3_arm_controller.
- Commented out the complete joint_state_publisher block.
- Removed the joint-state remapping from joint_states to franka/joint_states.

2.1 Import block

```
from launch import LaunchDescription
from launch.actions import (
    DeclareLaunchArgument,
    ExecuteProcess,
    IncludeLaunchDescription,
    Shutdown,
)
```

2.2 Controller startup order

The order was changed from:

```
for controller in ['fr3_arm_controller', 'joint_state_broadcaster']:
```

to:

```
for controller in ['joint_state_broadcaster', 'fr3_arm_controller']:
```

2.3 Rebuild after editing

```
cd ~/franka_ros2_ws
colcon build --packages-select franka_fr3_moveit_config
source ~/franka_ros2_ws/install/setup.bash
```

3. move_group.launch.py vs moveit.launch.py

The notes make an important distinction between the two launch files.

File	Role
move_group.launch.py	MoveIt backend only: loads the robot model, semantic model, kinematics, planner/contr
moveit.launch.py	Full MoveIt setup: MoveIt backend plus visualization, controllers, and supporting nodes.

3.1 What move_group.launch.py does not start

- RViz
- robot_state_publisher
- ros2_control_node
- joint_state_broadcaster
- fr3_arm_controller
- gripper launch

3.2 When to use move_group.launch.py

- When you want only the MoveIt backend.
- When another launch file will include MoveIt.
- When you want to connect your own RViz setup.
- When you want to connect your own control or simulation environment.

NOTE: Simple mental model: move_group.launch.py = the MoveIt brain only; moveit.launch.py = the MoveIt brain + visualization + controllers + support nodes.

4. Gazebo Startup and Launch-File Discovery

4.1 Start Gazebo with joint-position control and gripper

```
ros2 launch franka_gazebo_bringup \
  gazebo_joint_position_controller_example.launch.py \
  load_gripper:=true \
  franka_hand:=franka_hand
```

4.2 Find available Gazebo launch files

```
ls $(ros2 pkg prefix franka_gazebo_bringup)/share/franka_gazebo_bringup/launch
```

The corresponding source launch files are located in:

```
~/franka_ros2_ws/src/franka_gazebo/franka_gazebo_bringup/launch
```

4.3 Check trajectory-related topics

```
ros2 topic list | grep joint_trajectory
```

5. Gazebo Controller Examples

5.1 Joint Velocity Controller

Build the example-controller package first:

```
colcon build --packages-select franka_example_controllers
```

Then launch:

```
ros2 launch franka_gazebo_bringup \
  gazebo_joint_velocity_controller_example.launch.py \
  load_gripper:=true \
  franka_hand:=franka_hand
```

5.2 Joint Impedance Controller

Build the required ROS 2 control integration:

```
colcon build --packages-select franka_ign_ros2_control
source install/setup.sh
```

Then run the impedance example:

```
ros2 launch franka_gazebo_bringup \
  gazebo_joint_impedance_controller_example.launch.py \
  load_gripper:=true \
  franka_hand:=franka_hand
```

5.3 Joint Position Controller

Build the example controllers:

```
colcon build --packages-select franka_example_controllers
```

Launch with the gripper loaded:

```
ros2 launch franka_gazebo_bringup \
  gazebo_joint_position_controller_example.launch.py \
  load_gripper:=true
```

Launch with both gripper and Franka hand explicitly specified:

```
ros2 launch franka_gazebo_bringup \
  gazebo_joint_position_controller_example.launch.py \
  load_gripper:=true \
  franka_hand:=franka_hand
```

Complete version including the arm ID:

```
ros2 launch franka_gazebo_bringup \
```

```
gazebo_joint_position_controller_example.launch.py \
arm_id:=fr3 \
load_gripper:=true \
franka_hand:=franka_hand
```

6. Gripper Visualization and Motion Test

6.1 Terminal 1 - Visualize the Franka robot with gripper

```
ros2 launch franka_gazebo_bringup \
  visualize_franka_robot.launch.py \
  load_gripper:=true
```

6.2 Terminal 2 - Activate the gripper position controller

The notes state that fr3_gripper_position_controller was added to the file franka_gazebo_controllers.yaml under the franka_gazebo/franka_gazebo_bringup/config path.

```
ros2 control list_controllers

ros2 control load_controller --set-state active \
  fr3_gripper_position_controller
```

6.3 Run the custom gripper test

After the gripper controller is active, run the custom test_gripper.py script.

> NOTE: The supplied notes do not include the exact command or the contents of test_gripper.py, so this guide intentionally does not invent them.

7. Visualization-Only Launch

A direct visualization launch with the gripper enabled is:

```
ros2 launch franka_gazebo_bringup \
  visualize_franka_robot.launch.py \
  load_gripper:=true
```

8. Compact Workflow Reference

Goal	Main action
RViz fake hardware	Build franka_fr3_moveit_config, source workspace, launch moveit.launch.py with use_fake_
Show robot in RViz	RobotModel → Topic → /robot_description → Transient Local
MoveIt backend only	Use move_group.launch.py
Full MoveIt setup	Use moveit.launch.py
Gazebo joint position	gazebo_joint_position_controller_example.launch.py
Gazebo joint velocity	gazebo_joint_velocity_controller_example.launch.py
Gazebo joint impedance	gazebo_joint_impedance_controller_example.launch.py
Gripper visualization	visualize_franka_robot.launch.py load_gripper:=true
Gripper motion test	Load and activate fr3_gripper_position_controller, then run test_gripper.py

Prepared strictly from the three supplied note files. Repeated commands were consolidated, Greek comments were translated into English, and incomplete details were left explicitly incomplete rather than being filled in from outside sources.